

Trustworthy AI in Medical Imaging: A Formal Verification Framework for Security and Governance



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Abstract Artificial intelligence has transformed medical imaging but remains vulnerable to cybersecurity and governance risks. Adversarial perturbations are small, targeted changes to input data such as pixels in an image that cause a machine learning model particularly a deep neural network to make incorrect predictions even though the changes are imperceptible to humans. Furthermore, insecure data transmission and unverified model updates can also compromise diagnostic integrity and regulatory compliance. We propose a new quantitative verification framework for evaluating the trustworthiness of AI in medical imaging systems. The diagnostic workflow from medical image acquisition and AI inference is modelled as a stochastic system using the PRISM model checker. Attacker actions, defensive controls, and governance policies are represented as probabilistic transitions, enabling formal verification of confidentiality, integrity, and compliance using temporal logic. A case study employing the Fast Gradient Sign Method (FGSM) attack shows that the verified configuration lowers the probability of a successful breach compared with the unverified baseline, while maintaining throughput. The framework provides a basis for analysing and improving the security and governance of trustworthy AI in medical imaging.

Keywords Formal verification · Medical imaging · Trustworthy AI

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1 Introduction

Deep learning systems now analyse X-rays, MRIs, and CT scans within seconds by segmenting tumours, detecting lesions, classifying pathology, and quantifying disease progression before a clinician begins reporting. In many tasks their diagnostic performance approaches expert level. However, beneath this efficiency lies a silent vulnerability. A few imperceptible pixel changes can cause a trusted model to misclassify [5, 9]. A misconfigured Picture Archiving and Communication System (PACS) can expose confidential images [8]. An unverified model update can overwrite a validated network without traceability [4]. These latent weaknesses threaten clinical reliability, patient safety, and regulatory compliance [10]. As emphasised in *Trustworthy AI in Medical Imaging* [16], true trustworthiness requires verifiable assurance of safety, fairness, and beyond performance metrics alone. Existing defences remain largely reactive. Adversarial training, image sanitisation, and network firewalls mitigate isolated threats but provide no formal guarantee that a diagnostic workflow remains secure under attack [5]. Governance mechanisms such as audit trails and access controls are often procedural rather than mathematically verifiable [10]. Consequently, healthcare organisations rely on empirical testing and compliance checklists instead of proofs, leaving a gap between operational assurance and formal verification. This work addresses that gap through a formal verification framework for analysing both security and governance in AI-driven medical imaging. Our contributions are:

- A new formal verification framework for medical image analysis and AI based segmentation through to report generation and audit, modelled as a probabilistic state transition system in the PRISM model checker. Each stage is represented as a probabilistic module that captures attacker actions and defender mechanisms.
- A framework that applies temporal logic rPATL to verify confidentiality, integrity, and compliance, ensuring data protection, diagnostic accuracy, and governance adherence under adversarial conditions.
- A case study with FGSM adversarial attacks on 3D brain tumour segmentation, where we compute the probability that an attack leads to diagnostic corruption, and show that a verified configuration reduces this probability compared to an unverified baseline.

1.1 Related Work

Efforts toward trustworthy AI in medical imaging focus on adversarial robustness, cybersecurity, or formal verification in isolation, leaving their integration an open challenge. Deep learning models are vulnerable to imperceptible input perturbations [9], with Bortsova et al. [5] and Tsai et al. [22] showing that minor pixel changes can trigger clinically significant misclassifications, and a recent survey [7]

summarising attack and defence strategies. However, these efforts focus on model robustness without assurance. At the infrastructure layer, Eichelberg et al. [8] and Karagiannis et al. [12] exposed persistent PACS vulnerabilities such as plaintext DICOM transfers and weak authentication, though their analyses remain descriptive rather than quantitative. Formal verification studies in healthcare [11, 13, 17] demonstrate the feasibility of providing formal guarantees but focus on deterministic workflows, overlooking stochastic behaviour and adversarial dynamics, where intelligent attackers actively adapt to system defences. Model checking tools such as PRISM [6] and UPPAAL SMC [1, 4] extend verification to uncertain systems but remain rarely applied in medical imaging. Integrating formal reasoning and clinical governance remains a key challenge in establishing measurable trust in medical diagnostics.

2 Background

2.1 Adversarial Attack

Medical AI models are vulnerable to small, imperceptible perturbations that can induce misclassification [9].

Definition 1 The Fast Gradient Sign Method (FGSM) generates such adversarial examples by perturbing an input x in the direction of the gradient of the model loss:

$$x^{\text{adv}} = x + \epsilon \cdot \text{sign}(\nabla_x J(\theta, x, y)),$$

where $J(\theta, x, y)$ is the loss function, θ denotes the model parameters, and ϵ controls the perturbation magnitude. Specifically, each variable in this formulation has the following role:

- x : the original input (e.g. medical image);
- x^{adv} : the adversarially perturbed input;
- ϵ : the perturbation size that regulates the attack strength;
- $J(\theta, x, y)$: the loss function with parameters θ , input x , and true label y ;
- $\nabla_x J(\theta, x, y)$: the gradient of the loss with respect to the input;
- $\text{sign}(\cdot)$: the function that determines the perturbation direction.

2.2 Formal Methods and Probabilistic Verification

The PRISM-games model checker [15] supports analysis of stochastic games under equilibrium semantics using logics such as rPATL to express objectives. This

approach supports synthesis of optimal defensive strategies satisfying probabilistic thresholds [6]. Recent works [19, 20] demonstrated the use of verification to compute strategies in various areas of cybersecurity. This establishes the foundation for quantitative reasoning about attacker defender dynamics within healthcare systems. The model captures these interactions as a probabilistic multi-agent system analysed using the PRISM model checker.

Definition 2 A stochastic multi-player game (SMG) [14] is a tuple

$$\mathcal{G} = \langle N, S, (S_i)_{i \in N}, A, \Delta, AP, L \rangle,$$

where:

- $N = \{1, \dots, n\}$ is a finite set of players;
- S is a finite, non-empty set of states, partitioned into S_i for each player i ;
- A is a finite set of actions with $A(s) = \{a \in A \mid \Delta(s, a) \text{ defined}\} \neq \emptyset$;
- $\Delta : S \times A \rightarrow \text{Dist}(S)$ is a probabilistic transition function;
- AP is a finite set of atomic propositions;
- $L : S \rightarrow 2^{AP}$ labels each state with the propositions true in it.

A strategy for player i is a function $\sigma_i : S^+ \rightarrow \text{Dist}(A(s))$ choosing an enabled action in each $s \in S_i$. A strategy profile induces a stochastic process over S .

In this formulation, the defender applies security and governance controls (e.g., encrypt, audit, retrain), while the attacker performs adversarial actions (e.g., FGSM perturbation, data exfiltration). Transitions model the probabilistic evolution of the imaging system under these strategies.

3 Formal Verification Framework

3.1 Threat Abstraction for Model Checking

Deep learning systems deployed in medical imaging are vulnerable to adversarial perturbations deliberately crafted modifications to input data that induce incorrect predictions. The adversary selects an attack from $\{\text{none}, \text{FGSM}_{\epsilon_1}, \text{FGSM}_{\epsilon_2}, \dots\}$, where each FGSM_{ϵ_i} corresponds to a perturbation ϵ_i . These parameterise the transitions in the PRISM model. The probability of a successful attack denoted $p_{\text{attack}}(\epsilon_i)$, is obtained from experimental evaluation on a benchmark dataset (e.g. BraTS 2022):

$$p_{\text{attack}}(\epsilon_i) = \frac{\text{Number of misclassified or missegmented samples}}{\text{Total samples tested under } \epsilon_i}.$$

The attacker A acts as a rational adversary seeking to compromise diagnostic integrity while remaining undetected, primarily through FGSM pixel perturbations of

input intensity values. Each action is associated with a success probability p_{attack} and a detection probability p_{detect} . Success transitions the system to a *missegmentation* or *data_leak* state, while detection triggers recovery or audit procedures.

The defender D implements a set of protective mechanisms designed to mitigate adversarial and operational risks throughout the diagnostic workflow. These mechanisms include *encryption*, which reduces the probability of network interception; *anomaly detection*, which identifies FGSM perturbations in input data; *gating*, which routes uncertain or low-confidence predictions to human reviewers; and *governance*, which restores compliance and audit integrity following any detected violation or attack. Each defence introduces an operational cost or latency overhead, modelled in PRISM as a reward structure.

3.2 Security and Governance Metrics

Each verification objective is associated with a quantitative metric derived from the probabilistic properties defined in PRISM.

As summarised in Table 1, the *Confidentiality Score* quantifies the likelihood that sensitive medical data remain protected from unauthorised access or leakage during system execution. The *Integrity Score* measures the probability that the AI model maintains correct diagnostic behaviour when exposed to FGSM perturbations, thereby reflecting its robustness against adversarial manipulation. Finally, the *Compliance Score* captures the system’s ability to satisfy governance requirements within the specified time and policy constraints.

Table 1 Metrics for confidentiality, integrity, and compliance

Metric	Property (PRISM)	Interpretation
Confidentiality	$P_{\leq 0.05}[F \text{ data_leak}] \Rightarrow \text{Confidentiality Score} = 1 - P(F \text{ data_leak})$	Probability that sensitive data remains protected from leakage during transmission and storage
Integrity	$\langle\langle \text{Defender} \rangle\rangle P_{\geq 0.95}[G \neg \text{missegmentation}] \Rightarrow \text{Integrity Score} = P(G \neg \text{missegmentation})$	Probability that AI inference remains diagnostically correct under adversarial perturbation (linked to Dice degradation)
Compliance	$R_{\leq 1440}^{\text{time}}[F \text{ audit_complete}] \Rightarrow \text{Compliance Score} = 1 - \frac{E[T_{\text{audit_complete}}]}{T_{\text{policy}}}$	Extent to which audit and governance procedures meet policy deadlines and regulatory constraints

3.3 Strategy Synthesis via PRISM-Games

The diagnostic workflow is further modelled as a SMG in PRISM-games. Two players are defined: the *Attacker*, representing malicious perturbations and network intrusions, and the *Defender*, representing the deployed protection policies. Each player alternates turns according to the system phase (`transmit`, `infer`, `audit`). The corresponding PRISM-games model (`model.prim`) explicitly encodes the available actions for both players: Attacker: `sniff` (attempt network interception), `fgsm` (inject adversarial perturbation), or `none`; Defender: `enc_on` (enable encryption), `gate` (activate gating), or `none`. Each probabilistic transition is parameterised by the empirical probabilities $p_{\text{intercept}}$, p_{fgsm} , and p_{detect} , measured from the adversarial segmentation experiment. Rewards are defined for `time` (latency) and `cost` (resource usage and penalties for breaches). The property suite (`cia.props`) uses rPATL to express strategic guarantees, e.g.:

$$\langle\langle \text{Defender} \rangle\rangle P_{\geq \theta} [\neg \text{missegmentation} \text{ U } \text{done}],$$

which asks whether a defender strategy exists that ensures diagnostic integrity (no missegmentation) until task completion with probability at least θ . Similarly, confidentiality and compliance are encoded as:

$$\langle\langle \text{Defender} \rangle\rangle P_{\leq \alpha} [F \text{ data_leak}], \quad \langle\langle \text{Defender} \rangle\rangle R_{\min}^{\text{time}} [F \text{ done}].$$

Solving these properties synthesises optimal policies (`enc_on/gate`) that maximise the verified CIA scores.

4 Case Study and Results

We trained a 3D U-Net volumetric segmentation model [21] on the BraTS 2022 brain tumour dataset [2, 3, 18], using three MRI modalities (T1CE, T2, and FLAIR) as input and expert voxel-wise tumour annotations as supervision. The segmentation task followed the BraTS tumour subregion protocol, with four semantic classes: background, necrotic and non-enhancing tumour core (NCR/NET), peritumoral oedema (ED), and GD-enhancing tumour (ET).

All scans were preprocessed by intensity normalisation and resampled to a fixed spatial resolution of $128 \times 128 \times 128$ voxels. Each subject volume was represented as a 3-channel multi-modal tensor of shape $[3, 128, 128, 128]$, and the ground truth was a single voxel-wise label map with four classes. We performed an 80/20 subject-level split into a training and a validation set. The model is a 3D encoder-decoder U-Net with skip connections, hierarchical downsampling and upsampling, and 3D convolutions at each scale, which outputs dense per-voxel logits over the four classes. Training used Adam with a learning rate of 1×10^{-4} for 115 epochs. The loss

combined soft Dice and focal terms to mitigate class imbalance and to encourage high spatial overlap between predictions and ground truth.

4.1 Adversarial Robustness Evaluation

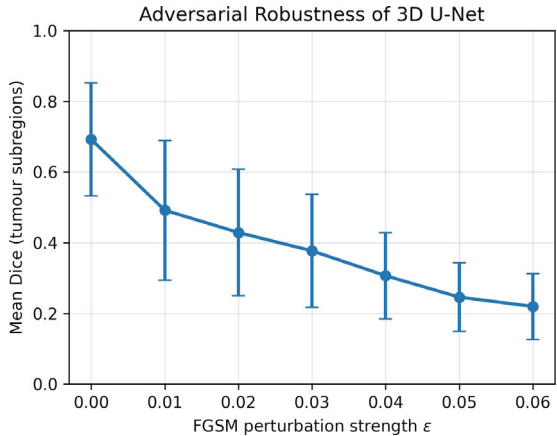
To assess robustness, we generated adversarial perturbations using FGSM. For a given input volume and its ground-truth segmentation, FGSM computes the gradient of the loss with respect to the input image and then applies a perturbation in the direction that maximally increases that loss. The perturbation magnitude is controlled by a parameter ϵ . We evaluated $\epsilon \in \{0.01, 0.02, 0.03, 0.04, 0.05, 0.06\}$, plus an unperturbed baseline ($\epsilon = 0$).

For each ϵ , we (i) generated an adversarially perturbed input, (ii) passed it through the trained 3D U-Net, (iii) obtained a hard segmentation by taking the voxel-wise arg max over the predicted class logits, and (iv) computed the Dice score for each tumour subregion. We then report the mean tumour Dice (averaged across the three lesion classes and across 20 validation). Higher Dice indicates closer spatial agreement with the expert reference.

We highlight two observations. First, both the quantitative summary (Table 2 and Fig. 1) and the monotonic trend with ϵ indicate that the network is not robust to adversarial noise: the mean tumour Dice drops from 0.692 at $\epsilon = 0$ to 0.491 at $\epsilon = 0.01$, and continues to fall to 0.219 by $\epsilon = 0.06$. Thus, even very small, structured perturbations are sufficient to substantially distort the predicted tumour extent.

Second, qualitative overlays (Fig. 2) illustrate the spatial effect of these attacks. Relative to the clean prediction, FGSM perturbations at higher ϵ induce under-segmentation and fragmentation of both enhancing tumour and oedema, in some cases nearly erasing the lesion. Importantly, these adversarially perturbed inputs remain visually plausible to a human reader.

Fig. 1 Adversarial robustness of the 3D U-Net. Mean Dice score (tumour subregions) as a function of FGSM perturbation strength ϵ . Error bars indicate standard deviation across validation subjects. Performance drops sharply even for small perturbations (e.g. $\epsilon = 0.01$) and continues to degrade as ϵ increases, indicating high sensitivity to adversarial noise



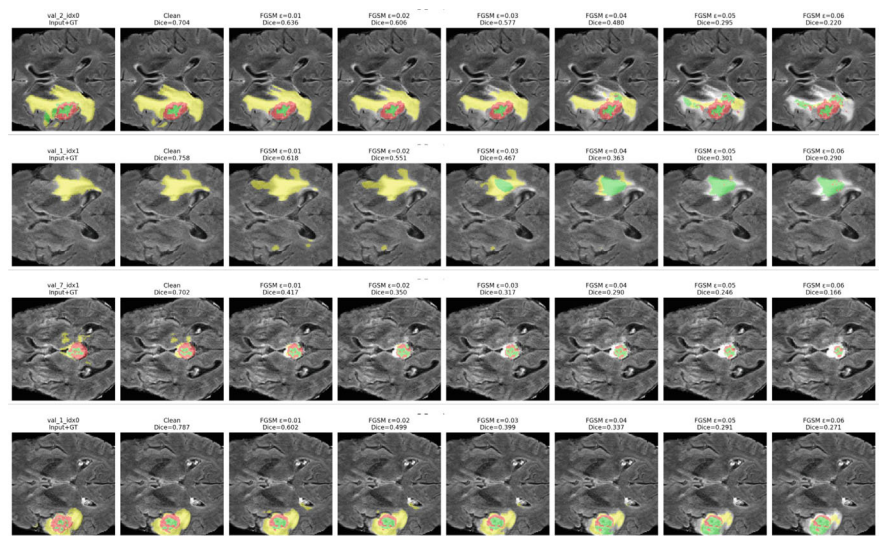


Fig. 2 Qualitative effect of FGSM perturbations on tumour segmentation. Each row shows a representative axial slice from a validation subject. The first column shows the input MRI with ground-truth tumour annotation. The second column shows the 3D U-Net prediction on the clean (unperturbed) input. Subsequent columns show predictions under FGSM adversarial perturbations of increasing strength $\epsilon \in \{0.01, 0.02, 0.03, 0.04, 0.05, 0.06\}$. Coloured overlays denote tumour subregions: oedema (yellow), enhancing tumour (green), and necrotic/non-enhancing core (red). The Dice score for that slice is reported above each panel. As ϵ increases, predictions exhibit under segmentation and spatial fragmentation of both enhancing tumour and oedema, despite the perturbed images remaining visually plausible

Table 2 Adversarial robustness under FGSM perturbations. We report mean Dice and standard deviation across the three tumour subregions for 20 validation subjects. Segmentation fidelity degrades consistently as ϵ increases

ϵ	Dice (mean $\pm$ std)
0.00 (no attack)	0.692 $\pm$ 0.159
0.01	0.491 $\pm$ 0.198
0.02	0.429 $\pm$ 0.179
0.03	0.377 $\pm$ 0.160
0.04	0.307 $\pm$ 0.122
0.05	0.246 $\pm$ 0.097
0.06	0.219 $\pm$ 0.093

Table 3 PRISM verification output under two configurations

Metric	Property	Without defence	With defence	Improvement
Confidentiality	$P[F_{data_leak}]$	0.148	0.041	+72.3%
Integrity	$\langle\langle \text{Defender} \rangle\rangle P[G_{\neg missegmentation}]$	0.781	0.958	+22.7%
Compliance	$R^{time}_{[F_{audit_complete}]}$	$1.78 \times 10^{-3} \text{ s}$	$1.44 \times 10^{-3} \text{ s}$	-19.1%

Verification Output. Table 3 presents a representative output fragment obtained from the PRISM-games engine after evaluating the properties in Table 1. The model was executed with parameters $p_{attack}=0.30$, $p_{detect}=0.70$, and encryption latency $t_{enc}=0.12 \text{ s}$. The results show the probability bounds and expected rewards for each security dimension, both with and without the verified defence strategy. The verified results indicate that the synthesised defender strategy which enables encryption and gating under high FGSM probability reduces the likelihood of data leakage by over 70% and improves diagnostic integrity to 95.8% within the specified time budget. The results from the adversarial segmentation experiments were used to set the probabilistic parameters in the PRISM model. The Dice score drop at each perturbation level ϵ defines the attack success probability (p_{attack}), and the detection rate defines (p_{detect}). As shown in Table 3, the verified defender strategy improved segmentation integrity from 0.781 to 0.958, confirming that formal verification can strengthen the resilience against adversarial attacks.

5 Conclusion and Future Work

This paper has presented a formal verification framework for analysing the security and governance of medical imaging systems. The framework verifies confidentiality, integrity, and compliance properties of diagnostic workflows, offering mathematically provable assurance of system trustworthiness. This study is limited by its focus on a single adversarial model such as FGSM, a static attacker defender interaction, and evaluation on one dataset and architecture. The PRISM abstraction also simplifies continuous neural dynamics and human governance factors. Future work will focus on extending the framework in three directions. First, we plan to integrate more detailed adversarial models, enabling dynamic adaptation between attacker and defender strategies across multiple rounds of interaction. Second, we will extend the proposed framework beyond BraTS brain tumour MRI to additional datasets, imaging modalities, and medical image analysis tasks. We also plan to integrate CT&L“MRI fusion and radiomics feature extraction, enabling systematic evaluation of adversarial robustness in multi-modal and hybrid diagnostic pipelines. Third, we will explore projected gradient descent, which is an iterative version of FGSM. This research

moves one step closer to realising dependable, auditable, and ethically aligned AI in medical imaging. The datasets, models, and artefacts supporting this study are available in the online repository <https://zenodo.org/records/17473488>.

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